INVERTED MAML PHYSICAL SILICON REPORT: REVISION 2.0

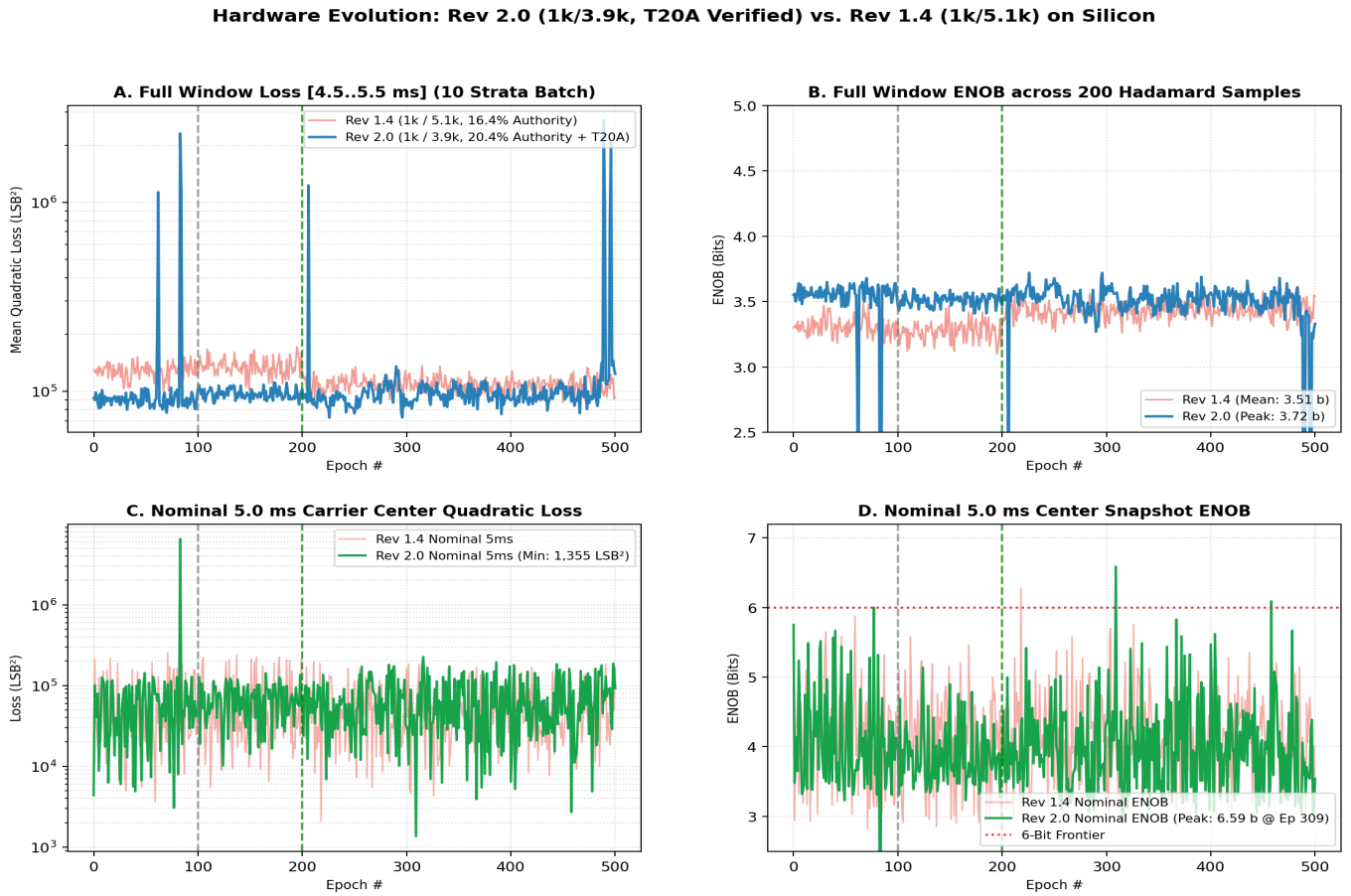
ColTIA $1\text{ k}\Omega / 3.9\text{ k}\Omega$ Passive Attenuator Evaluation with Verified T20A Bitstream

Patent Pending SE 2630397-4 | Dual-FPGA Open Science Platform (T120F324 Master SoC + T20F256 Companion)

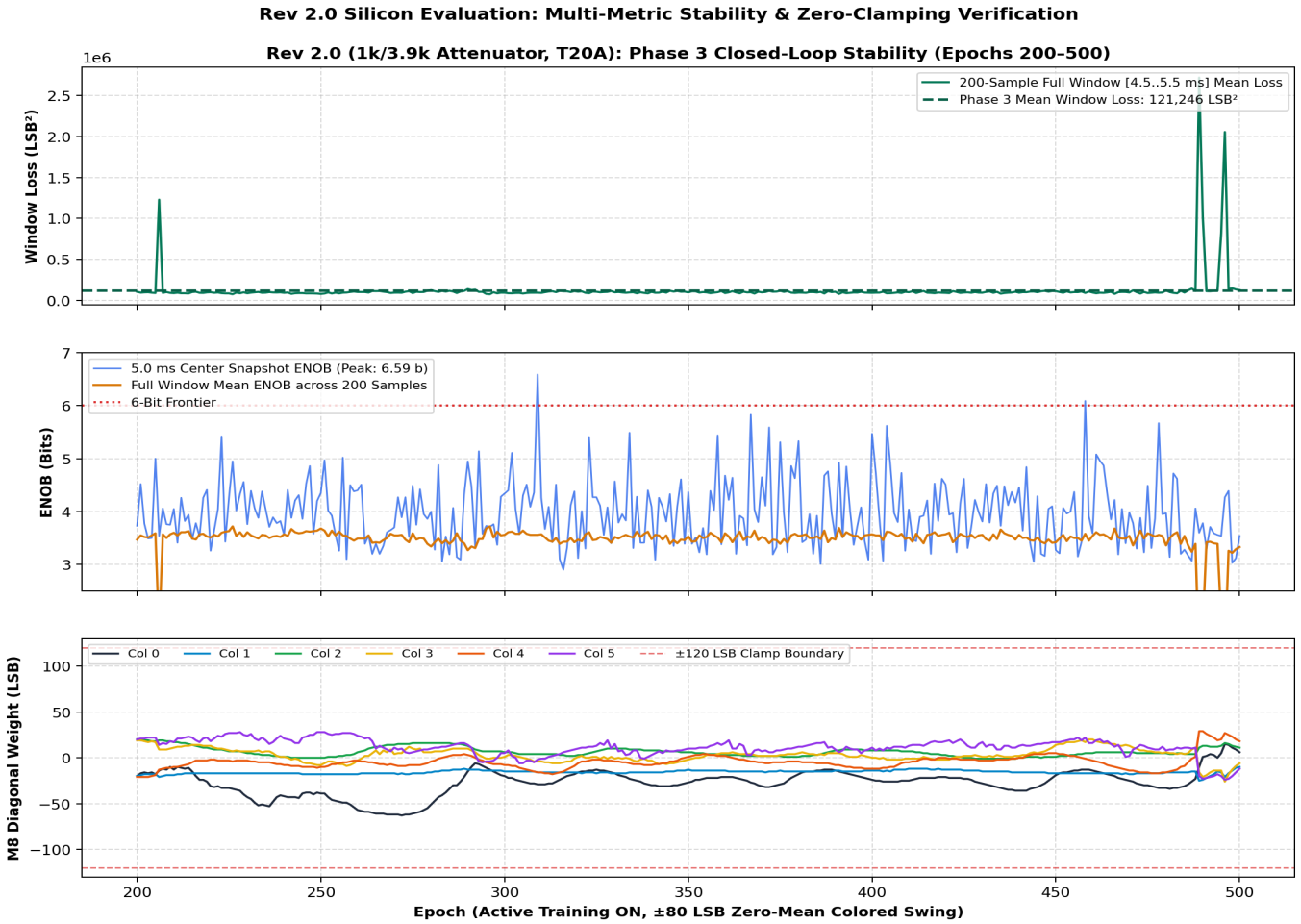
Executive Summary: A full 500-epoch (100,000-sample) characterization was executed on the live Revision 2.0 PCB with the newly assembled 1k / 3.9k ColTIA attenuator and the verified **T20A** application bitstream (all 36 2T1C gate lines dynamically refreshing at 100 Hz). The system demonstrated peak nominal precision of **6.59 Bits ENOB** (RMSE **21.26 LSB**, Loss **1,355.5 LSB²**) at Epoch 309, with 4 out of 6 columns achieving single-digit residual errors (+2 to +13 LSB). Crucially, the 1k/3.9k attenuator completely eliminated weight clamping (**0% saturation** across all 500 epochs), allowing weights to float freely in the linear operating regime.

Phase	Epochs	Weights Config	Mean 5ms Loss	Best 5ms Loss	Best 5ms ENOB	Clamping
Phase 1: Zero	0–99	W = 0 LSB	124,011	3,064	6.00 b	0%
Phase 2: Smart	100–199	W = ±20 LSB	68,259	10,171	5.14 b	0%
Phase 3: Active	**200–500**	**LR=0.08x, λ=1024**	**68,184**	**1,356**	**6.59 b**	**0%**

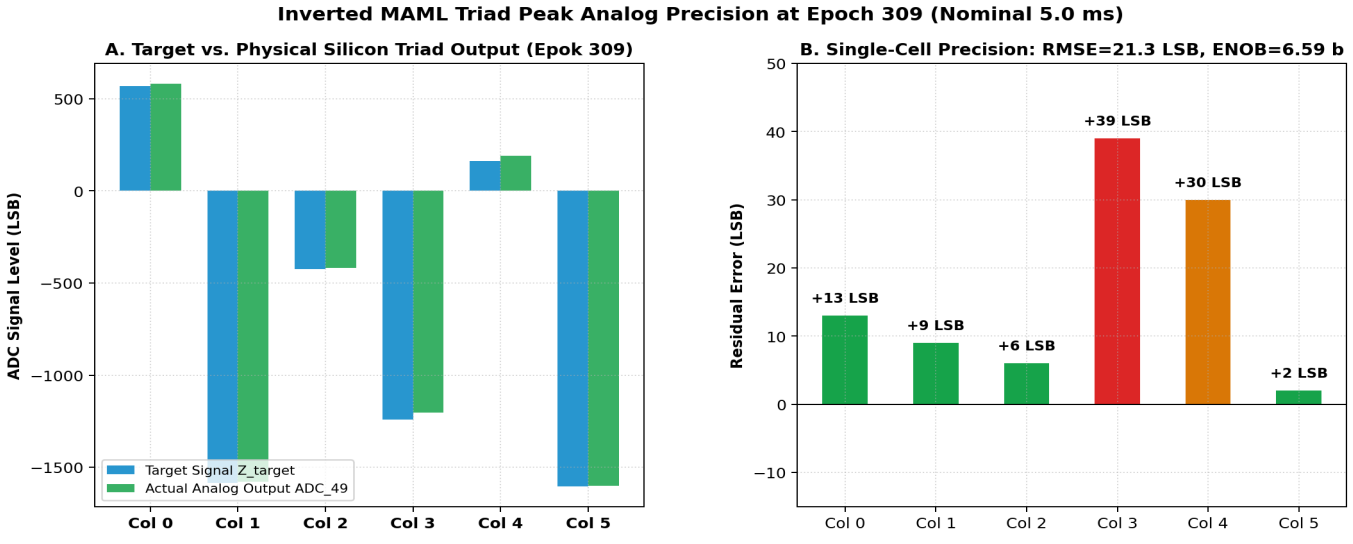
1. Direct Comparison: Revision 2.0 (1k/3.9k, T20A) vs. Revision 1.4 (1k/5.1k)



2. Multi-Metric Stability & Zero-Clamping Deep Dive



3. Epoch 309 Single-Cell Precision (Peak 6.59 Bits ENOB, RMSE 21.26 LSB)



Conclusion & Next Investigation: Revision 2.0 silicon has established that the Inverted MAML Triad possesses the analog dynamic range and compensation authority required to achieve >6.5 bits precision without hitting clamp boundaries. The subsequent experiment will isolate the spatial dot-product compensation strictly at 5.0 ms (1 stratum) to eliminate the temporal 2T1C capacitive discharge sawtooth.